

HAT-P-55b

R-0067

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-55_250225 · created 2026-07-15T15:54:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460731.9703 +/- 0.0087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.024
Transit depth (Rp/Rs)^2	1.76 +/- 0.64 [%]
Orbital Inclination (inc)	85.78 +/- 1.7
Airmass coefficient 1 (a1)	1.034 +/- 0.014
Airmass coefficient 2 (a2)	-0.026 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#2 - [361, 132]
Optimal Aperture	2.24
Optimal Annulus	11.45
Transit Duration (day)	0.103 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.024 vs 0.1202 (deviation 0.53σ)
Duration fit vs published	0.103 d vs 0.1225 d (deviation 0.72σ)
Mid-time O–C	23.1 min
Depth SNR	5.5
Residual scatter	1.31 %

Light curve

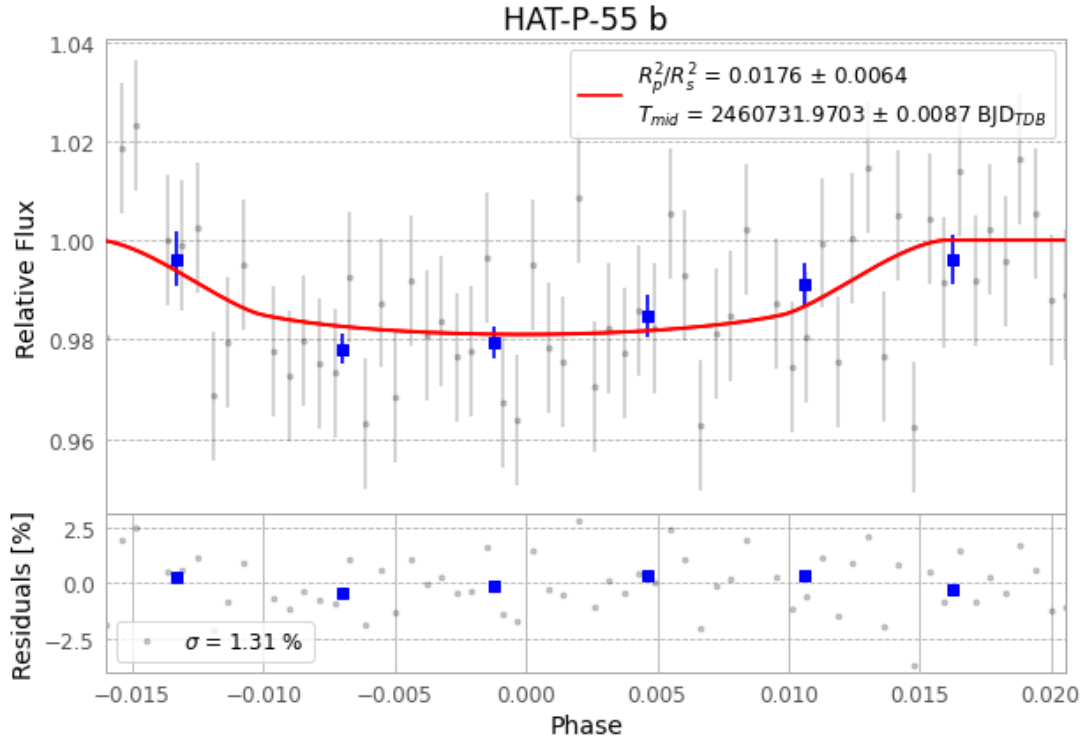


Figure 1 — FinalLightCurve_HAT-P-55 b_2025-02-25.png (SHA-256 1a0a6da213f09ae1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c9d960e92ede6fd0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

64 FITS frames (42 MB), HATP-55250225095416.FITS → HATP-55250225130315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-250225.log (8f987ab4d5a2...), FinalLightCurve_HAT-P-55 b_2025-02-25.png (1a0a6da213f0...), FinalLightCurve_HAT-P-55 b_2025-02-25.csv (15cd0d970e36...)

Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.